MFE FORMWORK MR11 SYSTEM

API Reference

Version 1.0

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1. Introduction

This reference describes the system's API for anyone who needs to call it directly, for example to automate an upload or to read the MR11 into another tool. The API has 24 endpoints under `/api`, and they all exchange JSON over HTTPS.

Apart from sign-in and the two health checks, every endpoint needs a session token. You get the token by signing in, and you send it with each request in the header `Authorization: Bearer <token>`.

2. Conventions

- **Base address:** `https://<host>/api`
- **Request bodies:** JSON (`Content-Type: application/json`), except for uploads, which are sent as a multipart form.
- **Successful responses** look like `{ "success": true, "data": ... }`.
- **Errors** look like `{ "success": false, "error": { "code": "...", "message": "..." } }`.
- **Department codes:** `BD`, `FINANCE`, `SHELLPLAN`, `DESIGN`, `PLANNING`, `PRODUCTION` and `DISPATCH`. `SHELLPLAN` and `DESIGN` both return the same combined workbook.
- **Size limits:** JSON bodies can be up to 100 MB and uploaded files up to 25 MB.

3. Signing in and roles

1. Call `POST /api/auth/login` with an email address and password.
2. Keep the `token` you get back. It is a JWT signed with the server's `JWT_SECRET`, and it is valid for 7 days.
3. Send it with every request as `Authorization: Bearer <token>`.
4. The token carries the roles the user had when they signed in, so after a role change the user needs to sign in again to get a new token.

Role	What it can call
Any signed-in user	`/auth/me`, `/auth/change-password`, the `/departments` list, and every MR11 read, sync and export endpoint
A department role	That department's own `/departments/:code` endpoints (Shell Plan and Design share theirs)
CEO	`/visualization`
ADMIN	Everything, including `/admin/*` and deleting or restoring MR11 rows

4. Error codes

HTTP status	Code	When it happens
400	`VALIDATION_ERROR`	The email, password or password fields are missing
400	`FILE_REQUIRED`	An upload was sent without a file

HTTP status	Code	When it happens
400	WEAK_PASSWORD, SAME_PASSWORD, DEFAULT_PASSWORD, WRONG_PASSWORD	A password change broke one of the rules
401	AUTH_REQUIRED	No Bearer token was sent
401	INVALID_TOKEN	The token has expired, is invalid, or was signed with a different secret
401	USER_NOT_FOUND, WRONG_PASSWORD	Sign-in failed
403	ACCESS_DENIED, UNAUTHORIZED_DEPARTMENT_UPLOAD	The user's role is not allowed to do this
404	DEPT_NOT_FOUND, FILE_NOT_FOUND	The department is unknown, or no original file is stored
500	UPLOAD_FAILED, SAVE_FAILED, AUTH_ERROR	Something went wrong on the server or in the database; the message explains what

5. Health checks

Method	Path	Access	What it returns
GET	/health	Public	{ status: "ok", service: "MFE Formwork MR11 System" }
GET	/api/health/db	Public	A status of CONNECTED, NOT_CONFIGURED or ERROR, the connection string with the password hidden, the server time, the PostgreSQL version, the list of tables, and the active files and row counts

6. Authentication endpoints (/api/auth)

Method	Path	Access	Body	What it returns
POST	/login	Public	{ email, password }	The token and user { id, email, fullName, roles[], usingDefaultPassword }, also repeated under data. Errors: 400 VALIDATION_ERROR, 401 USER_NOT_FOUND, 401 WRONG_PASSWORD
GET	/me	Signed in	None	data.user { id, email, fullName, roles[] }, or 404 if the account no longer exists
POST	/change-password	Signed in	{ currentPassword, newPassword }	{ success: true, message: "Password changed." }. Errors: 400 WEAK_PASSWORD (under 8 characters), SAME_PASSWORD, DEFAULT_PASSWORD or WRONG_PASSWORD, and 500 SAVE_FAILED

7. Department endpoints (/api/departments)

Method	Path	Access	Input	What it returns
GET	/	Signed in	None	Every department, with `activeVersion { id, originalFilename, status, uploadedAt }`
GET	`/:code` (also `/:code/workbook` and `/:code/active`)	That department, or ADMIN	None	The department with its `activeVersion`, including the `parsedWorkbook`
POST	`/:code/upload` (also `POST /:code`)	That department, or ADMIN	A multipart form with the field `file` (.xlsx, up to 25 MB). For Dispatch only, add `part=local` or `part=overseas` as a query or form field; if you leave it out, the server works it out from the file name or columns	"<code> workbook uploaded and set as active. Master MR11 regenerated." Errors: 400 `FILE_REQUIRED`, 403, and 500 `UPLOAD_FAILED`
GET	`/:code/download`	That department, or ADMIN	None	The original file, or 404 `FILE_NOT_FOUND` for files uploaded through the browser (only the parsed data is kept)

8. MR11 endpoints (/api/mr11)

Method	Path	Access	Input	What it returns
GET	/	Signed in	None	See section 8.1
POST	`/regenerate`	Signed in	None	Rebuilds the MR11 from every department's latest workbook and returns `{ runId }`
GET	`/export`	Signed in	None	The file `MR11_Master_Order_<timestamp>.xlsx`, or 400 if there is no MR11 yet
GET	`/planning-series`	Signed in	None	`[{ projectNo, projectShortname, stream, fontColor, seriesNumber, totalProcessed, totalQuantity, closingDate }]`
GET	`/production-series`	Signed in	None	`[{ projectShortname, stream, fontColor, seriesNumber, totalProduced }]`
GET	`/shellplan-history`	Signed in	None	`[{ projectNo, shortName, projectName, stream, fillColor, status, revision, approvedDate, submittedDate, recordedOn }]`, newest first
GET	`/dispatch-monthly`	Signed in	None	`[{ shortName, projectNo, stream, productType, file, totalDispatch, months: { "2026-10": m² } }]`
GET	`/deleted-rows`	ADMIN	None	`[{ key, shortName, projectNo, projectName, stream, productType, deletedAt, deletedBy }]`

Method	Path	Access	Input	What it returns
POST	/rows/delete	ADMIN	{ key }, the row's _rowKey	The updated list of deleted rows, after the MR11 is rebuilt. Returns 404 if the row is no longer in the MR11
POST	/rows/restore	ADMIN	{ key }	The updated list of deleted rows, after the MR11 is rebuilt

8.1 What GET /api/mr11 returns

Field	What it contains
run	{ id, status, generatedAt, recordCount, records[], sourceSnapshot }, with the prices in records worked out for today
fxRate	{ rate, source, asOf, live }, the USD to MYR rate
currencyRates	[{ code, rate, inBd, source }], the value of 1 USD in each currency shown
lmePrice	{ cash, threeMonth, previousCash, asOf, source, live }, in USD per tonne
orderedHeaders	The 78 MR11 column names, in the order they are shown
headerGroups	The groups shown under a two-row header (Payment terms)
numberFormats	The number of decimal places for each column
columnDepartments and departmentColors	Which department each column comes from, and that department's highlight colour

9. Administration endpoints (/api/admin, ADMIN only)

Method	Path	Body	What it returns
GET	/users	None	[{ id, email, fullName, createdAt, roles[] }]
POST	/users	{ email, password, fullName, primaryProfile }, where the profile is ADMIN, BD, FINANCE, SHELLPLAN, DESIGN, PLANNING, PRODUCTION or DISPATCH	201 with the new user, or 400 if the email is already in use
PATCH	/users/:userId/permissions	{ roles: ["BD", ...] } or { primaryProfile }	The user with their new roles. The main admin account always keeps ADMIN
DELETE	/users/:userId	None	Confirms the deletion, or 403 for the main admin account

10. Visualization endpoint (/api/visualization)

Method	Path	Access	What it returns
GET	/	ADMIN and CEO	`kpi { totalProjects, totalSellingUSD, totalOrderedM2, totalProduced }`, `designStatusCounts` and `streamDistribution`. Please note that `totalOrderedM2` and `totalProduced` always come back as 0 because of a known issue; the CEO Executive Hub uses `/api/mr11` instead

11. Examples

```
# sign in and keep the token
TOKEN=$(curl -s https://<host>/api/auth/login -H "Content-Type: application/json" \
  -d '{"email":"user@example.com","password":"<password>"}' | jq -r .token)

# read the MR11
curl -s https://<host>/api/mr11 -H "Authorization: Bearer $TOKEN"

# upload the BD workbook
curl -s -X POST https://<host>/api/departments/BD/upload -H "Authorization: Bearer $TOKEN" -F
"file=@bd.xlsx"

# upload the Overseas Dispatch file
curl -s -X POST "https://<host>/api/departments/DISPATCH/upload?part=overseas" -H
"Authorization: Bearer $TOKEN" -F "file=@dispatch-overseas.xlsx"

# export the MR11 to Excel
curl -s -o mr11.xlsx https://<host>/api/mr11/export -H "Authorization: Bearer $TOKEN"
```